In the late 1970s and early 1980s, Schwarz teamed up with Michael Green, then of Queen Mary College in London, and set to work on some of the technical hurdles facing string theory. Primary among these was the problem of *anomalies.* The details are not of the essence, but, roughly speaking, an anomaly is a pernicious quantum effect that spells doom for a theory by implying that it violates certain sacred principles, such as energy conservation. To be viable, a theory must be free of all anomalies. Initial investigations had revealed that string theory was plagued by anomalies, which was one of the main technical reasons it had failed to generate much enthusiasm. The anomalies signified that although string theory appeared to provide a quantum theory of gravity, since it contained gravitons, on closer inspection the theory suffered from its own subtle mathematical inconsistencies.

Schwarz realized, however, that the situation was not clear-cut. There was a chance—it was a long shot—that a complete calculation would reveal that the various quantum contributions to the anomalies afflicting string theory, when combined correctly, cancelled each other out. Together with Green, Schwarz undertook the arduous task of calculating these anomalies, and by the summer of 1984 the two hit pay dirt. One stormy night, while working late at the Aspen Center for Physics in Colorado, they completed one of the field's most important calculations—a calculation proving that all of the potential anomalies, in a way that seemed almost miraculous, *did* cancel each other out. String theory, they revealed, was free of anomalies and hence suffered from no mathematical inconsistencies. String theory, they demonstrated convincingly, was quantum mechanically viable.

This time physicists listened. It was the mid-1980s, and the climate in physics had shifted considerably. Many of the essential features of the three nongravitational forces had been worked out theoretically and confirmed experimentally. Although important details remained unresolved—and some still do—the community was ready to tackle the next major problem: the merging of general relativity and quantum mechanics. Then, out of a little-known corner of physics, Green and Schwarz burst on the scene with a definite, mathematically consistent, and aesthetically pleasing proposal for how to proceed. Almost overnight, the number of researchers working on string theory leaped from two to over a thousand. The first string revolution was under way.